Seat Booking Encapsulation Guard

Practice Problem 3

Concept and approach

Keep the total and available seat counts private. Reject zero or negative capacity during construction. Maintain the invariant 0 <= seatsAvailable <= seatsTotal; booking decrements only if a seat remains, and cancellation increments only if there is a booking to undo.

Important implementation notes

Construction with 0 throws IllegalArgumentException, so the program's caller can handle an invalid capacity. The invalid third booking/cancellation silently leaves the state unchanged.

Java solution

```
public class CineScreen {
private final int seatsTotal;
private int seatsAvailable;

public CineScreen(int seatsTotal) {
if (seatsTotal <= 0 || seatsTotal > 300)
throw new IllegalArgumentException("seatsTotal must be between 1 and 300");
this.seatsTotal = seatsTotal;
this.seatsAvailable = seatsTotal;
}

public void bookSeat() {
if (seatsAvailable > 0) seatsAvailable--;
}

public void cancelBooking() {
if (seatsAvailable < seatsTotal) seatsAvailable++;
}

public int getSeatsAvailable() { return seatsAvailable; }

public static void main(String[] args) {
CineScreen c = new CineScreen(2);
c.bookSeat(); c.bookSeat(); c.bookSeat();
System.out.println(c.getSeatsAvailable());
c.cancelBooking(); c.cancelBooking(); c.cancelBooking();
System.out.println(c.getSeatsAvailable());
}
}
```

Sample output

```
0
2
```